

Git Practice Project Report

Author: Oluwaseun Osunsola Environment: Hyper Terminal on Windows GitHub Link:

<https://github.com/OluwaseunOa/Cybersecurity-Projects/tree/main/Cyber-Security-Fundamentals>

Welcome to my **Git Practice** project! This report documents a hands-on tutorial walking through the complete Git workflow—from initializing a local repository, creating files, pushing to GitHub, branching, editing, creating pull requests, merging changes, and syncing back locally.

All steps are illustrated with terminal screenshots for clarity. Follow along to master Git fundamentals!

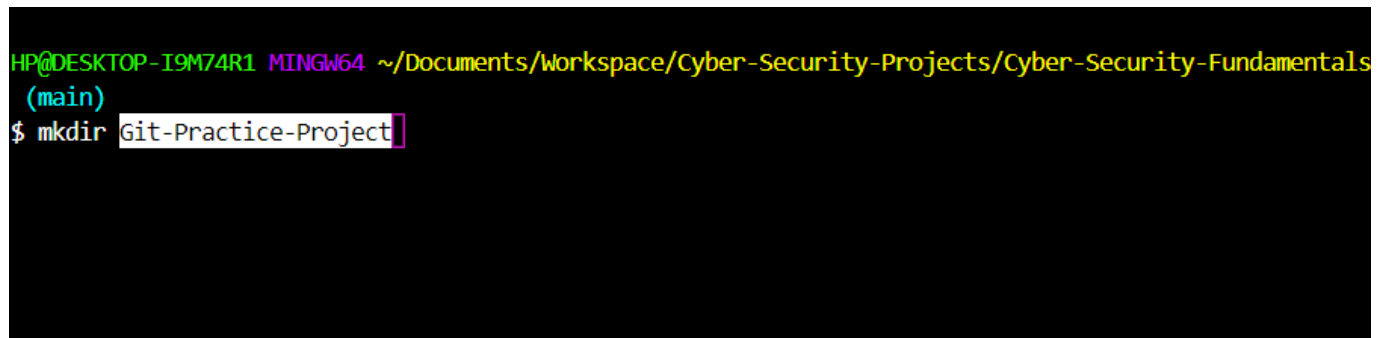
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1. Local Repository Setup

Step 1: Create Project Folder

Created the main project folder using Hyper Terminal.

A terminal window with a black background and green text. The prompt is 'HP@DESKTOP-I9M74R1 MINGW64 ~'. The current directory is '/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals'. The user is on the '(main)' branch. The command '\$ mkdir Git-Practice-Project' has been entered, and the folder name is highlighted with a white box.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals
(main)
$ mkdir Git-Practice-Project
```

Step 2: Navigate into Project Folder

`cd` into the newly created folder.

```
MINGW64:/c/Users/HP/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project
(main)
$ cd Git-Practice-Project/

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project
(main)
$
```

Step 3: Initialize Git Repository

Run `git init` to turn the folder into a Git repository.

```
MINGW64:/c/Users/HP/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/Git-Practice-Project
$ git init
Initialized empty Git repository in C:/Users/HP/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/Git-Practice-Project/.git/

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/Git-Practice-Project
(main)
$
```

Step 4: Create & Edit README.md

Created `README.md` file and added a heading.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$ echo "# Git Practice Project" > README.md

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$
```

Step 5: Verify README Creation

Used `ls` command to confirm `README.md` exists.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$ ls README.md
README.md

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$
```

Step 6: Check Git Status

Ran `git status` to see untracked files.

```
MINGW64:/c/Users/HP/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        img/

no changes added to commit (use "git add" and/or "git commit -a")

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$
```

Step 7: Stage All Changes

Used `git add .` to stage all changes.

```
MINGW64:/c/Users/HP/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$ git add .
warning: in the working copy of 'Cyber-Security-Fundamentals/Git-Practice-Project/README.md', LF will be replaced by CRLF the next time Git touches it

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$
```

Step 8: Commit Changes

Committed the staged `README.md` with a message.

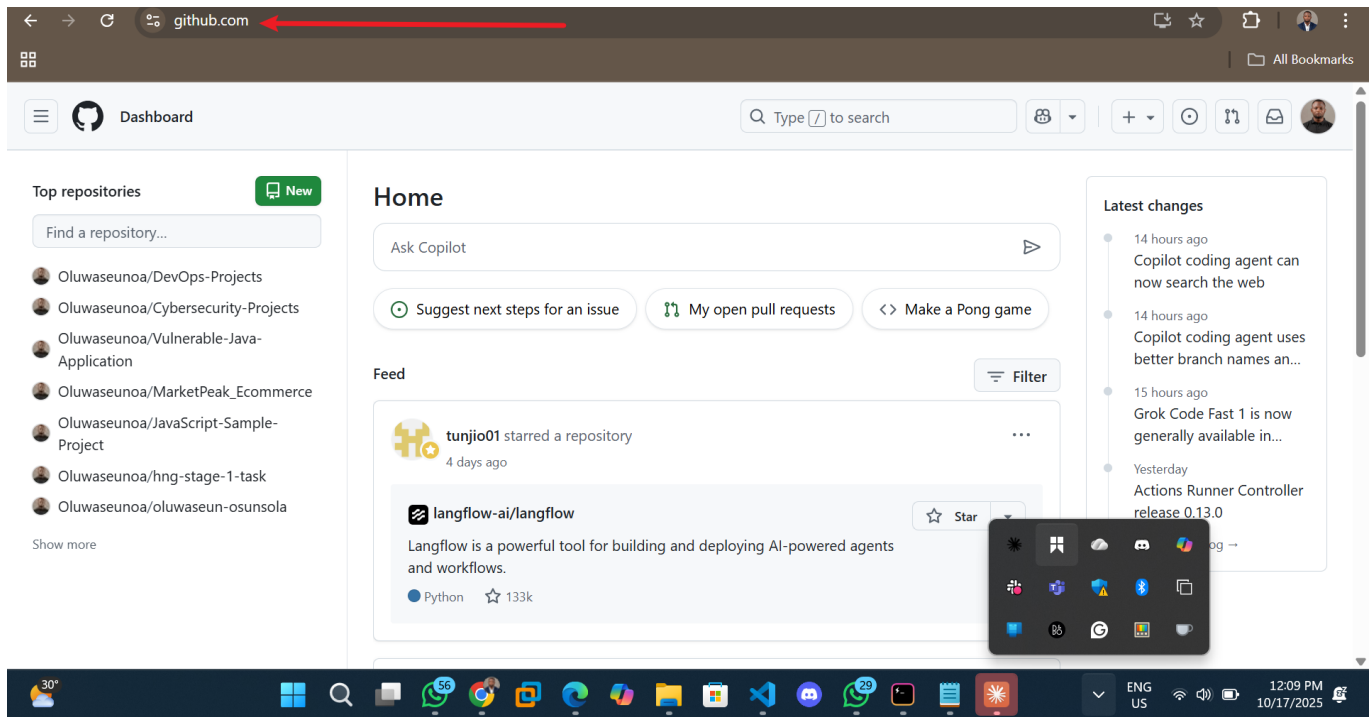
```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$ git commit -m "Initial commit of README with heading"
[main 4629063] Initial commit of README with heading
 6 files changed, 1 insertion(+)
   create mode 100644 Cyber-Security-Fundamentals/Git-Practice-Project/img/1.Created_Git-Practice-Project_folder_on_Hyper_Terminal.png
   create mode 100644 Cyber-Security-Fundamentals/Git-Practice-Project/img/2.Navigated_into_Git-Practice-Project_folder.png
   create mode 100644 Cyber-Security-Fundamentals/Git-Practice-Project/img/3.created_README_file_and_Added_heading_to_it.png
   create mode 100644 Cyber-Security-Fundamentals/Git-Practice-Project/img/4.confirm_that_README.md_was_created_with_ls_command.png
   create mode 100644 Cyber-Security-Fundamentals/Git-Practice-Project/img/5.ran_git-status_to_check_untracked_files.png

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cyber-Security-Projects/Cyber-Security-Fundamentals/Git-Practice-Project (main)
$
```

2. GitHub Remote Repository Creation

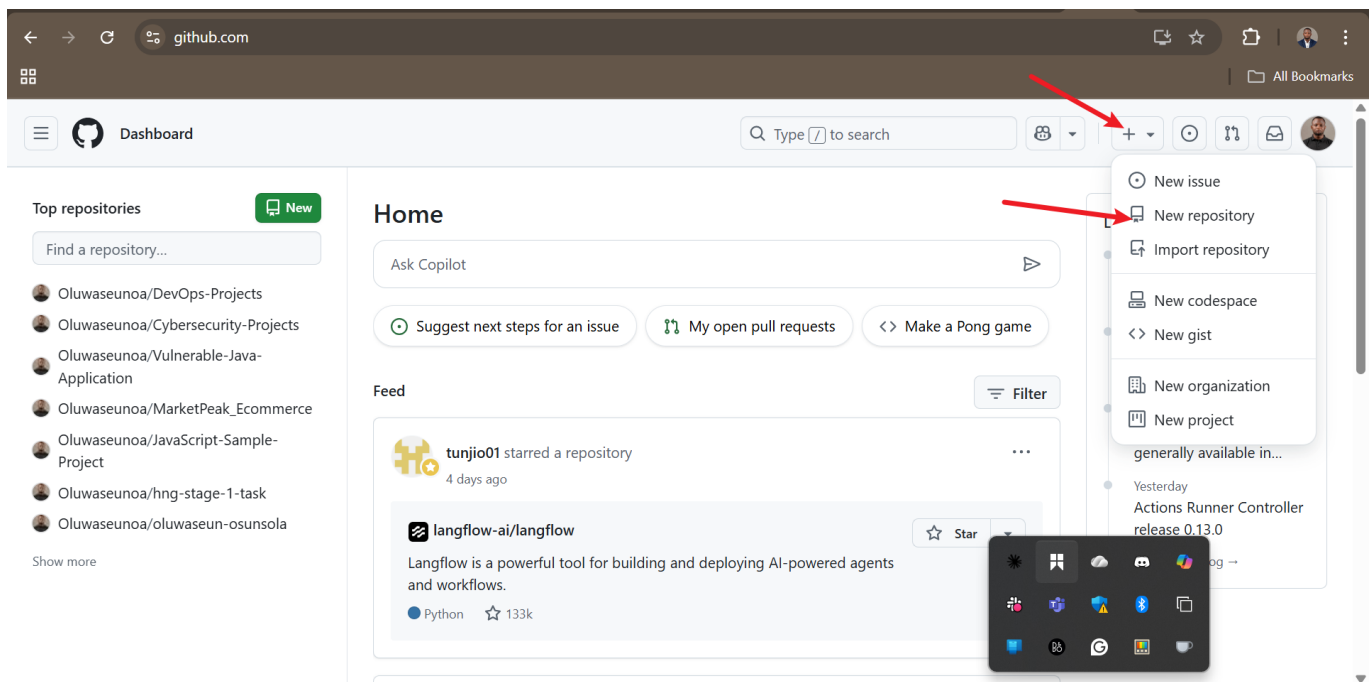
Step 9: Open GitHub

Navigated to github.com in the browser.



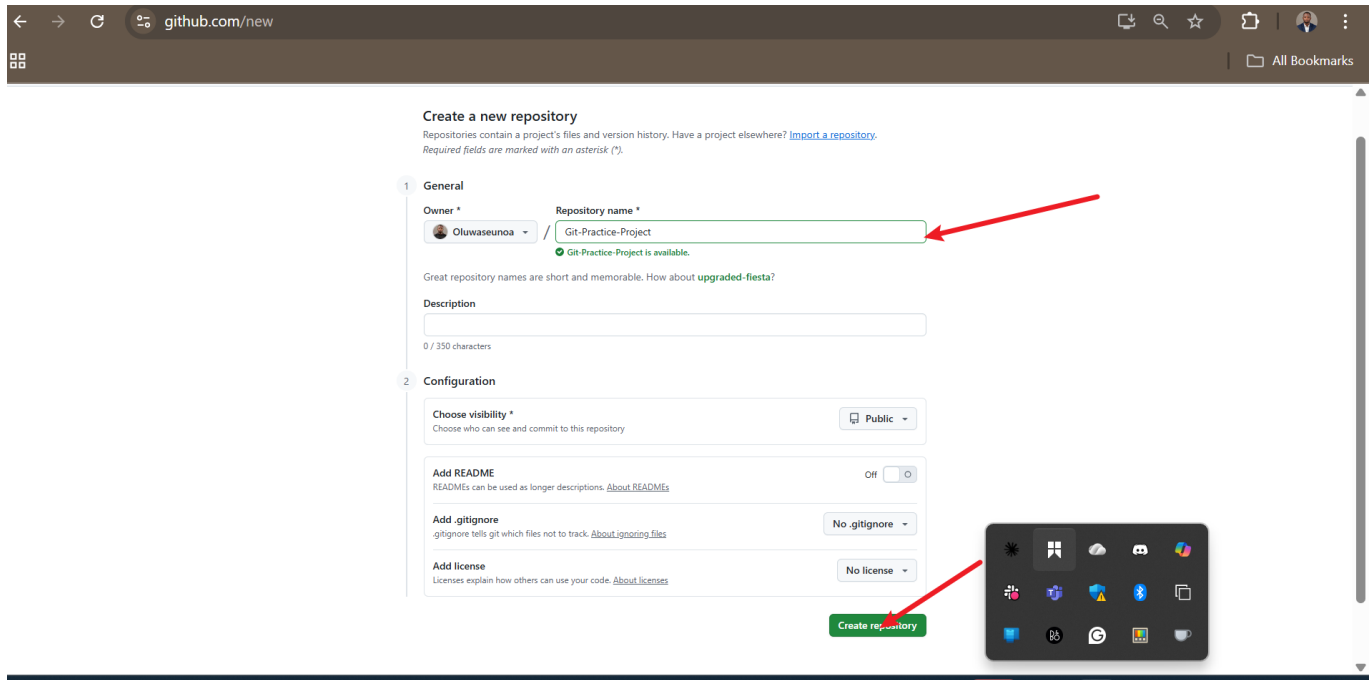
Step 10: Create New Repository

Clicked the **+** icon → **New Repository**.



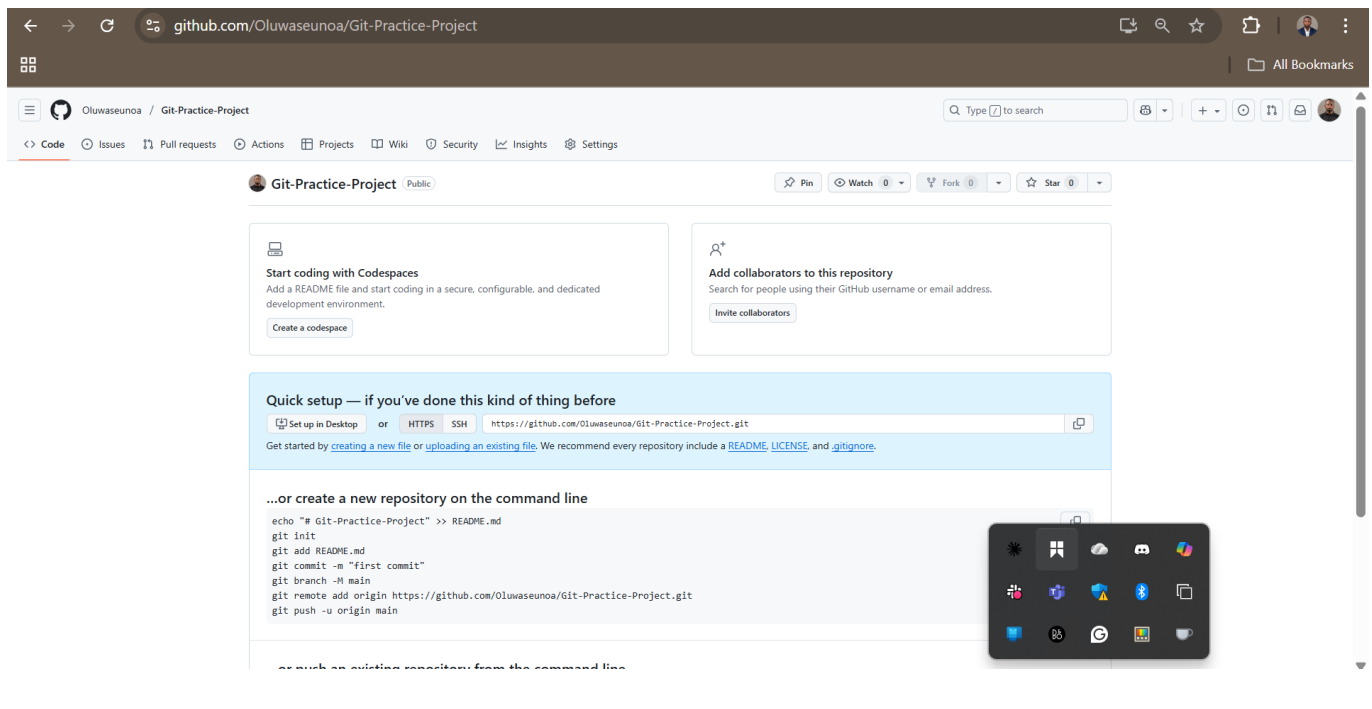
Step 11: Name & Create Repository

Entered repository name: **Git-Practice-Project** and clicked **Create Repository**.



Step 12: Repository Created Successfully

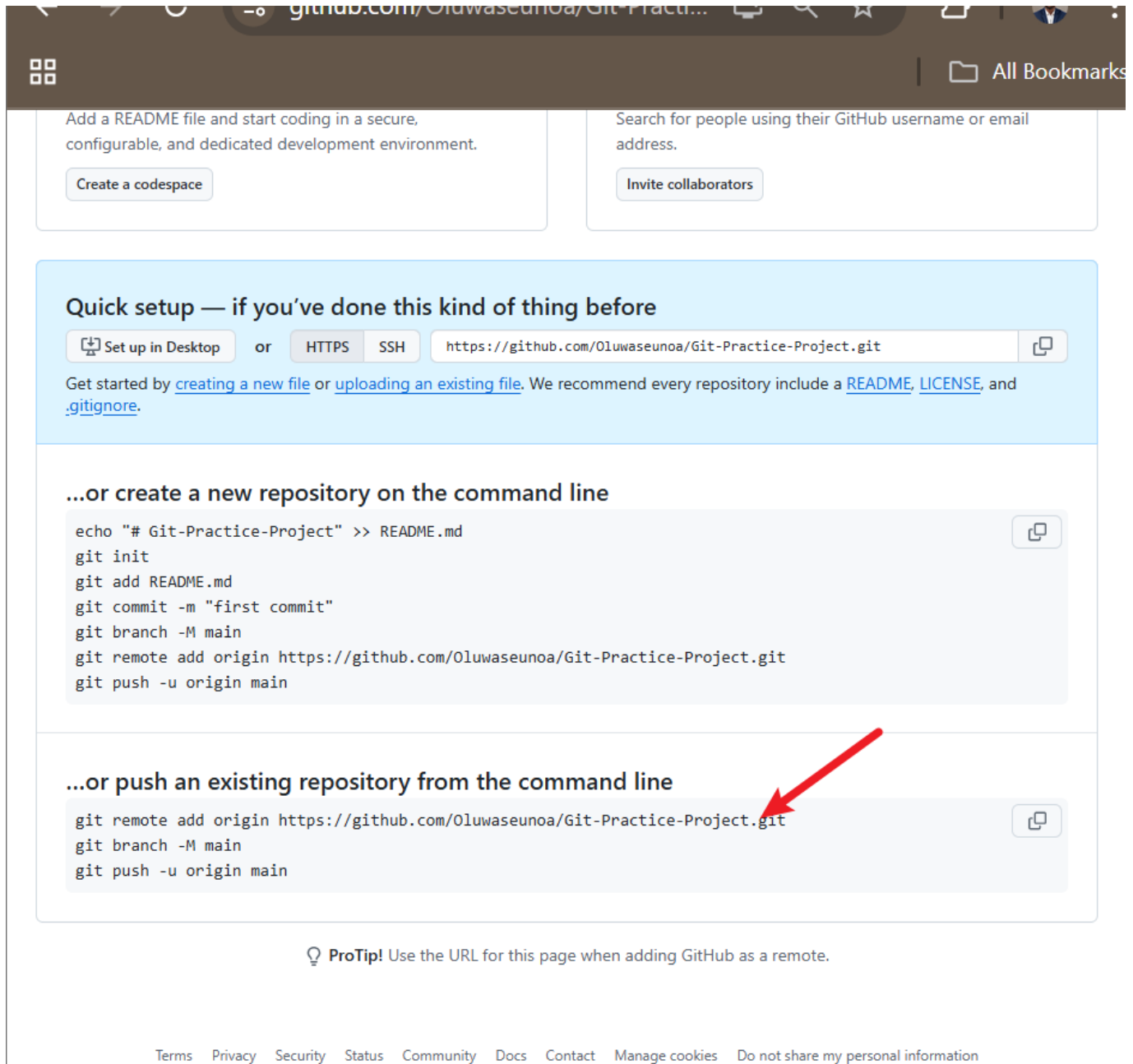
Empty repository ready on GitHub!



3. Linking & Pushing Local Repo to GitHub

Step 13: Copy Remote HTTPS Link

Copied the HTTPS link from GitHub repo page.



The screenshot shows the GitHub repository page for 'Git-Practice-Project'. At the top, there are two buttons: 'Create a codespace' and 'Invite collaborators'. Below these, a light blue box titled 'Quick setup — if you've done this kind of thing before' contains a 'Set up in Desktop' button, an 'or' separator, and 'HTTPS' and 'SSH' tabs. The 'HTTPS' tab is selected, showing the URL 'https://github.com/Oluwaseunola/Git-Practice-Project.git'. Below this, it says 'Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).' Below the blue box, there are two sections for command-line setup. The first section, '...or create a new repository on the command line', shows a series of commands: 'echo "# Git-Practice-Project" >> README.md', 'git init', 'git add README.md', 'git commit -m "first commit"', 'git branch -M main', 'git remote add origin https://github.com/Oluwaseunola/Git-Practice-Project.git', and 'git push -u origin main'. The second section, '...or push an existing repository from the command line', shows the commands: 'git remote add origin https://github.com/Oluwaseunola/Git-Practice-Project.git', 'git branch -M main', and 'git push -u origin main'. A red arrow points to the URL in the second section. At the bottom, a 'ProTip!' says 'Use the URL for this page when adding GitHub as a remote.' The footer contains links for Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

Add a README file and start coding in a secure, configurable, and dedicated development environment.

Create a codespace

Search for people using their GitHub username or email address.

Invite collaborators

Quick setup — if you've done this kind of thing before

 Set up in Desktop or **HTTPS** **SSH** `https://github.com/Oluwaseunola/Git-Practice-Project.git` 

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# Git-Practice-Project" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/Oluwaseunola/Git-Practice-Project.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/Oluwaseunola/Git-Practice-Project.git
git branch -M main
git push -u origin main
```

 **ProTip!** Use the URL for this page when adding GitHub as a remote.

[Terms](#) [Privacy](#) [Security](#) [Status](#) [Community](#) [Docs](#) [Contact](#) [Manage cookies](#) [Do not share my personal information](#)

Step 14: Link Local to Remote

Used `git remote add origin <URL>` with copied link.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$ git remote add origin https://github.com/Oluwaseunola/Git-Practice-Project.git

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$
```

Step 15: Verify Remote Link

Ran `git remote -v` to confirm connection.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$ git remote -v
origin https://github.com/Oluwaseunola/Git-Practice-Project.git (fetch)
origin https://github.com/Oluwaseunola/Git-Practice-Project.git (push)

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$
```

Step 16: Push to GitHub

Pushed local repo: `git push -u origin main`.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$ git push -u origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 8 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (15/15), 972.15 KiB | 3.47 MiB/s, done.
Total 15 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Oluwaseunola/Git-Practice-Project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/Cybersecurity-Projects/Cyber-Security-Fundamentals/
Git-Practice-Project (main)
$
```

4. Cloning Repository

Step 17: Create Clone Folder

Created `CloneRepoFolder` in Documents to clone into.


```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents
$ mkdir CloneRepoFolder
```

Step 17b: Navigate into Clone Folder

`cd CloneRepoFolder.`

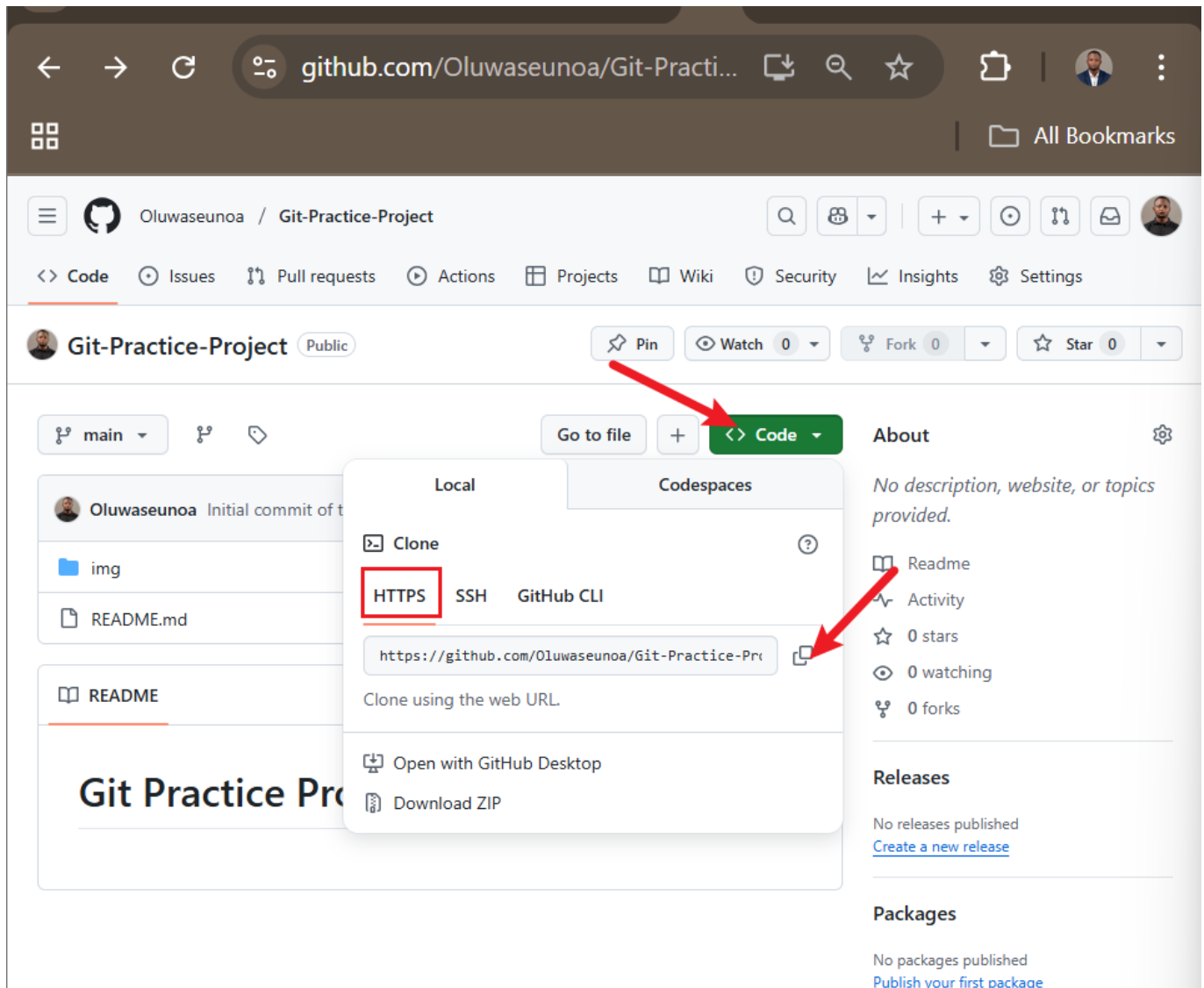
```
MINGW64:/c/Users/HP/Documents/CloneRepoFolder

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents
$ cd CloneRepoFolder

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder
$ cd
```

Step 18: Copy HTTPS Link Again

Clicked **Code** button → **HTTPS** → Copy link.



Step 19: Clone Repository

Ran `git clone <URL>` and `cd` into cloned folder.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder
$ git clone https://github.com/Oluwaseunola/Git-Practice-Project.git
Cloning into 'Git-Practice-Project'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 15 (delta 0), reused 15 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 972.15 KiB | 345.00 KiB/s, done.

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder
$ ls
Git-Practice-Project/

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder
$ cd Git-Practice-Project/

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$
```

```
MINGW64:/c/Users/HP/Documents/CloneRepoFolder/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder
$ cd Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$
```

5. Branching & Editing README

Step 20: Create Feature Branch

```
git checkout -b feature-edit-readme.
```

```
MINGW64:/c/Users/HP/Documents/CloneRepoFolder/Git-Practice-Project

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$ git checkout -b feature-edit-readme
Switched to a new branch 'feature-edit-readme'

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$
```

Step 21: Edit README.md

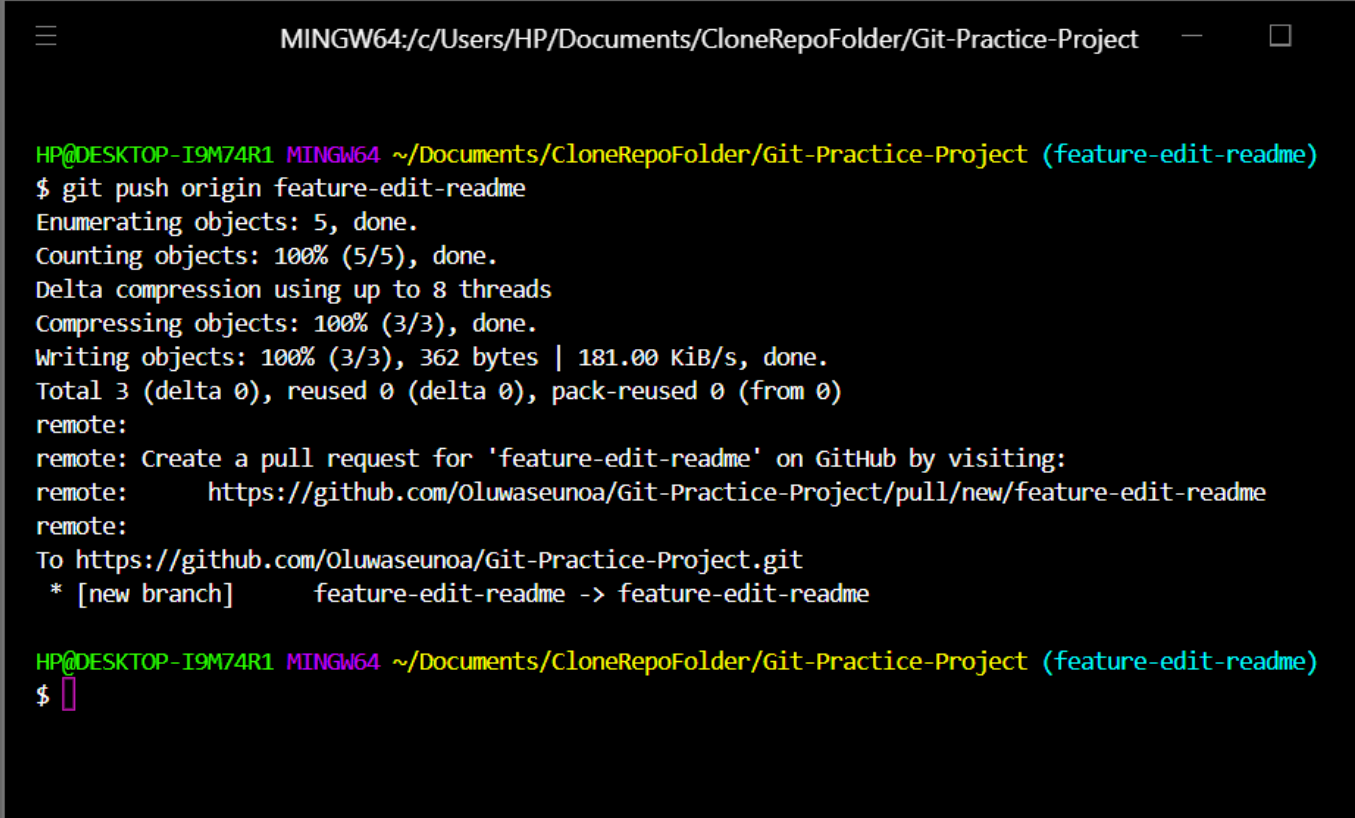
Added a new line to `README.md` on the feature branch.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$ echo "This was added from the feature branch" >> README.md

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$
```

Step 22: Stage, Commit & Push Branch

`git add .` → `git commit -m "..."` → `git push origin feature-edit-readme`.

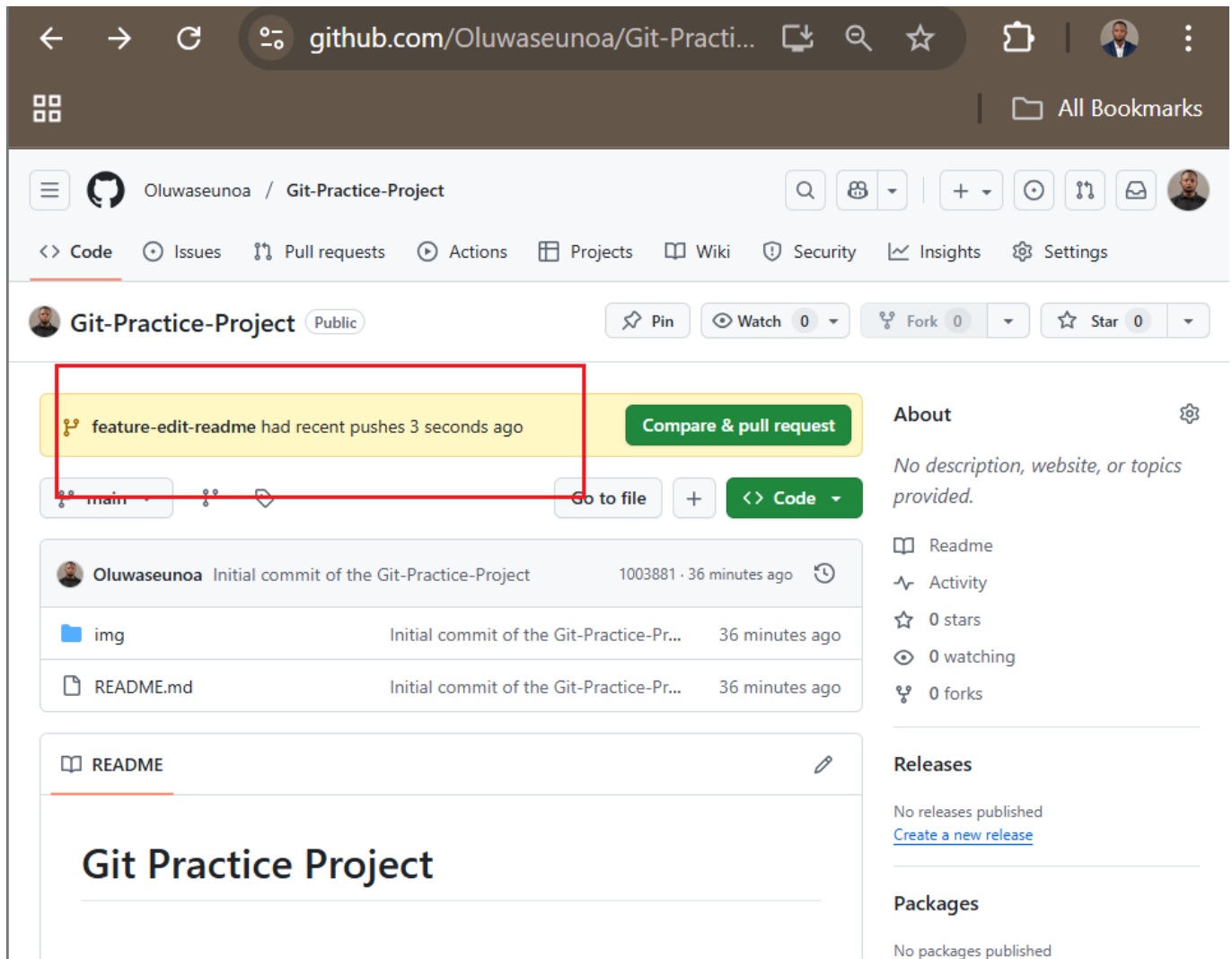
A terminal window with a black background and white text. The title bar at the top reads 'MINGW64:/c/Users/HP/Documents/CloneRepoFolder/Git-Practice-Project'. The prompt is 'HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)'. The command '\$ git push origin feature-edit-readme' has been executed. The output shows the progress of pushing the branch to the remote repository, including enumerating objects, counting objects, and compressing objects. It also provides instructions to create a pull request on GitHub.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$ git push origin feature-edit-readme
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 362 bytes | 181.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-edit-readme' on GitHub by visiting:
remote:   https://github.com/Oluwaseunola/Git-Practice-Project/pull/new/feature-edit-readme
remote:
To https://github.com/Oluwaseunola/Git-Practice-Project.git
 * [new branch]      feature-edit-readme -> feature-edit-readme

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$
```

Step 23: Branch Visible on GitHub

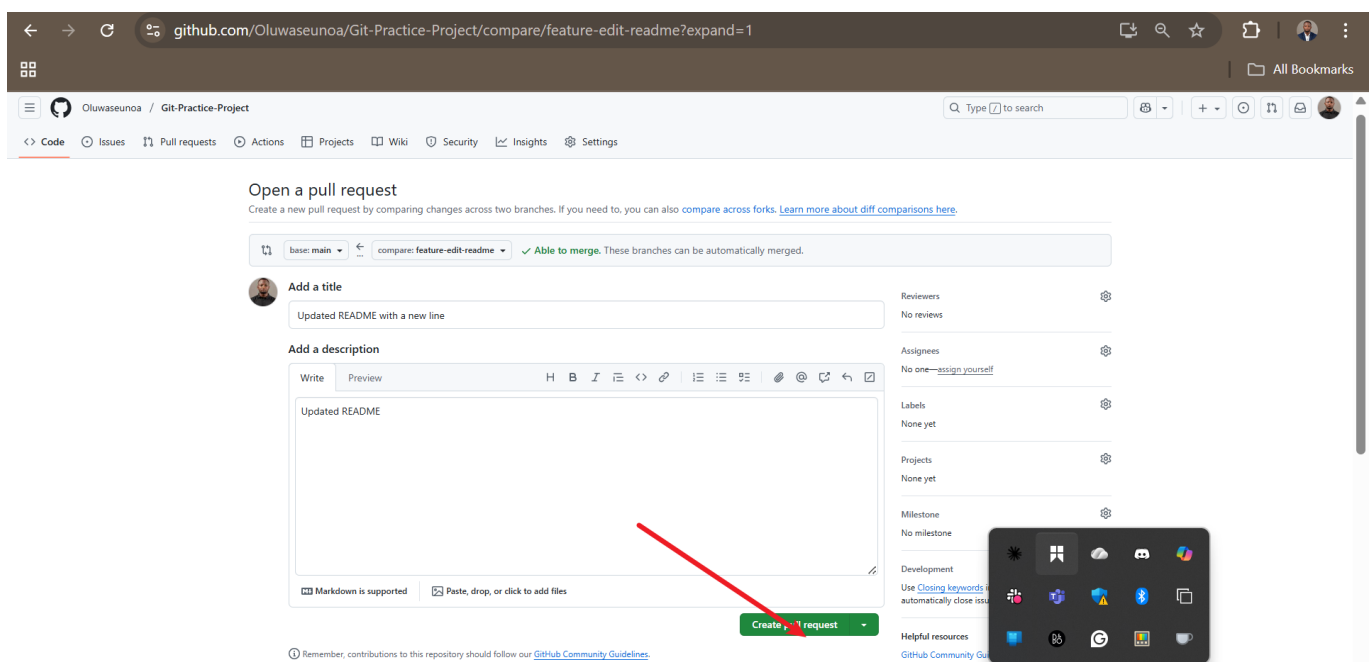
New branch successfully pushed!



6. Pull Request & Merge

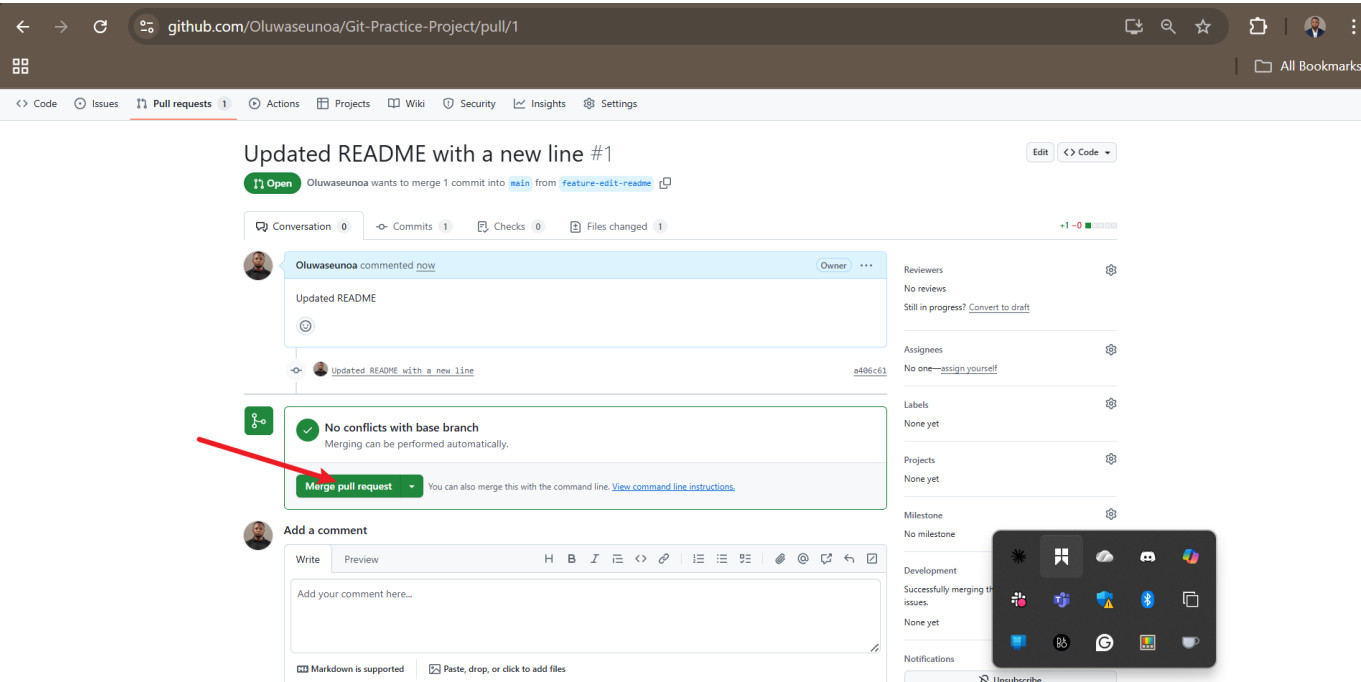
Step 24: Create Pull Request

Clicked **Compare & pull request**.



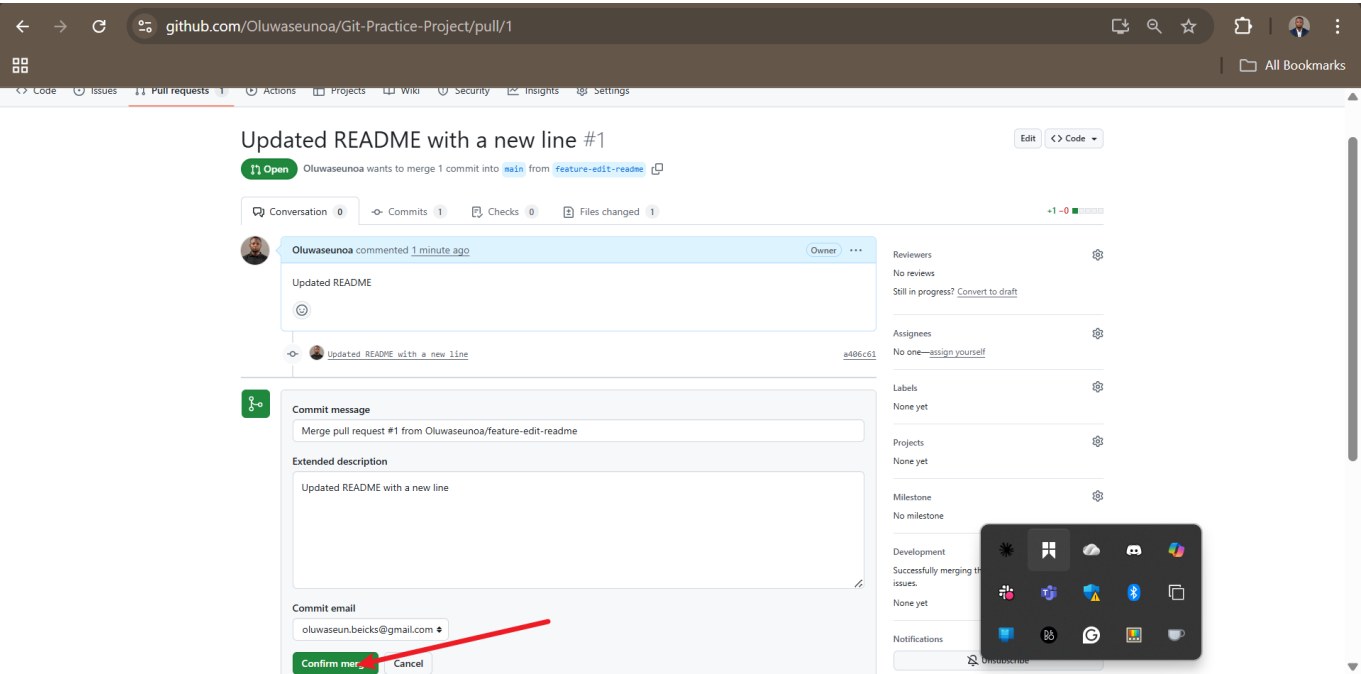
Step 25: Merge Pull Request

Reviewed changes and clicked **Merge pull request**.



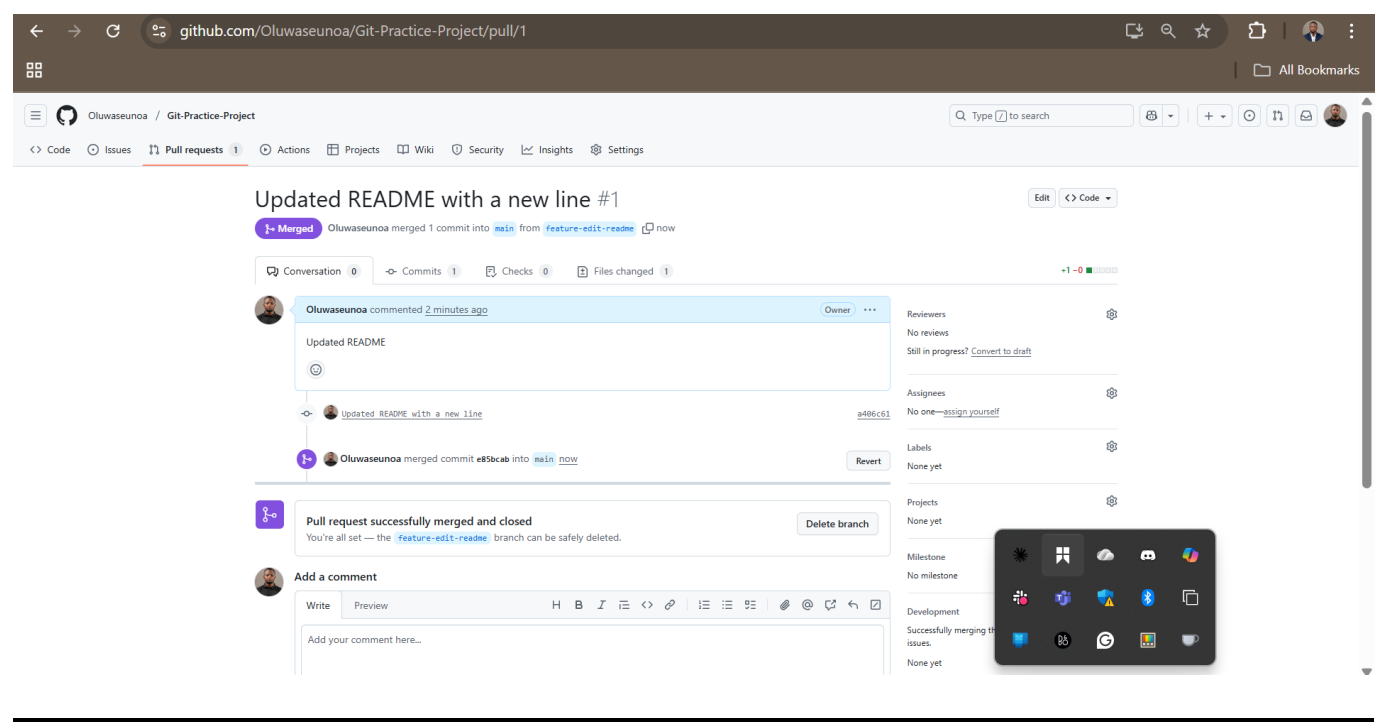
Step 26: Confirm Merge

Clicked **Confirm merge**.



Step 27: Merge Successful!

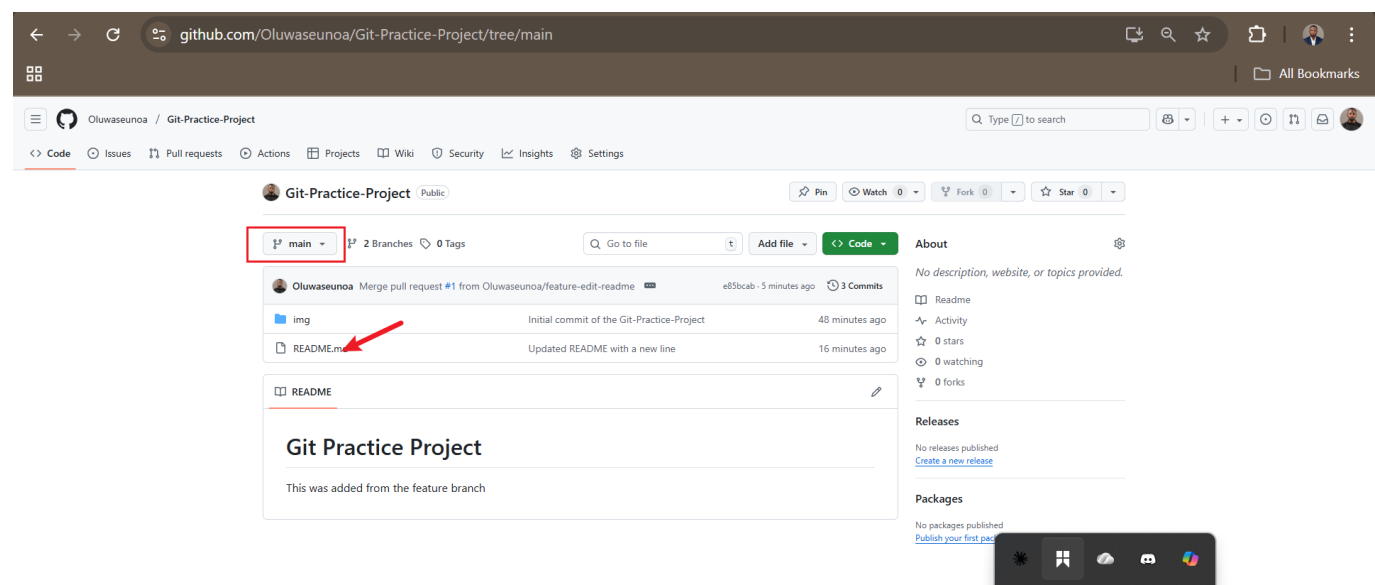
Pull request merged into **main** branch.



7. Verify Merged Changes

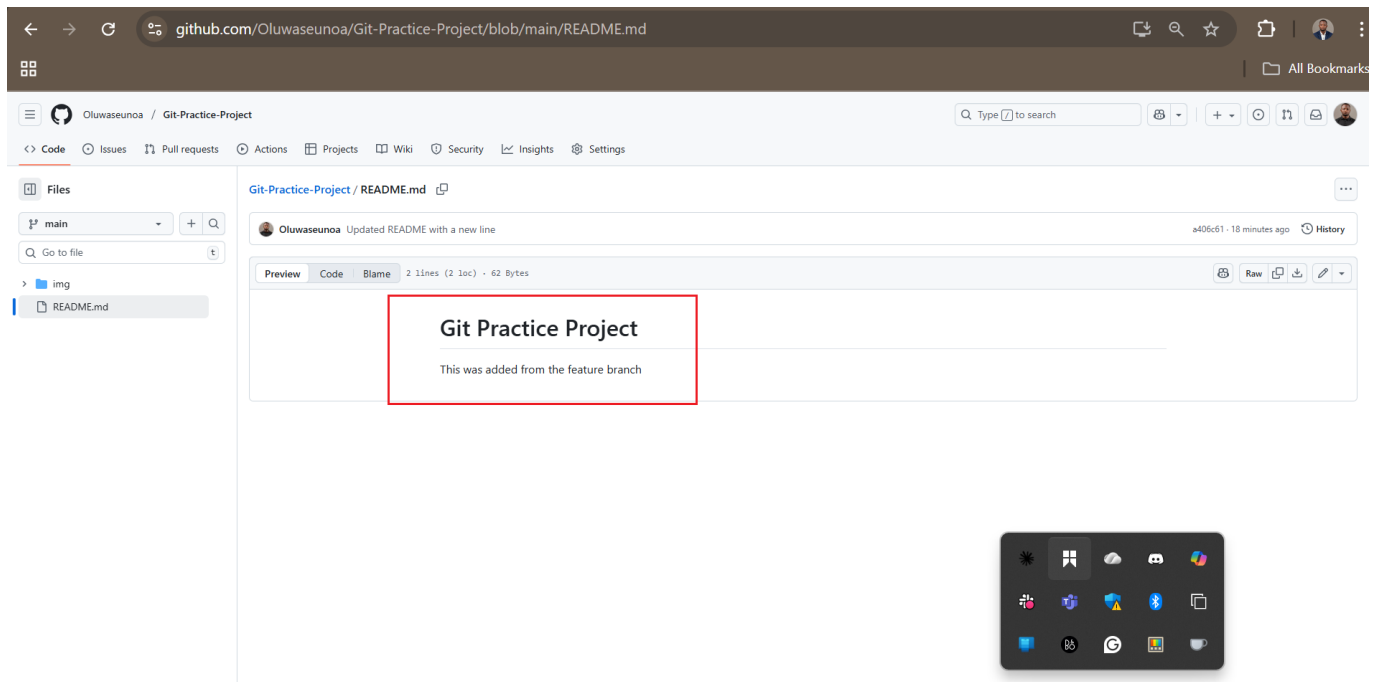
Step 28: Open README on GitHub

Navigated back to repo files and opened **README.md**.



Step 29: New Line Visible

The added line now appears in **main** branch README!



8. Sync Local Main Branch

Step 30: Checkout Main Locally

Switched to `main` branch—new line missing locally.

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (feature-edit-readme)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$ cat README.md
# Git Practice Project
```

Step 31: Pull Latest Changes

Ran `git pull origin main`—new line now added locally!

```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$ git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 912 bytes | 152.00 KiB/s, done.
From https://github.com/Oluwaseunola/Git-Practice-Project
   1003881..e85bcab  main       -> origin/main
Updating 1003881..e85bcab
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$ cat README.md
# Git Practice Project
This was added from the feature branch

HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/CloneRepoFolder/Git-Practice-Project (main)
$
```

Summary & Key Learnings

Skill Demonstrated	Commands Used
Initialize repo	<code>git init</code>
Create & stage files	<code>touch, git add .</code>
Commit changes	<code>git commit -m "..."</code>
Link to GitHub	<code>git remote add origin <URL></code>
Push to remote	<code>git push -u origin main</code>
Clone repo	<code>git clone <URL></code>
Create branch	<code>git checkout -b <branch></code>
Push branch	<code>git push origin <branch></code>
Pull changes	<code>git pull origin main</code>

Total Steps: 31

Time to Complete: ~30 minutes

Status: ☒ **Complete!**

This project proves I can confidently handle the **full Git workflow** end-to-end. Ready for collaborative coding! 

Authored on October 17, 2025 | [View on GitHub](#)